

Research Article

Impact of Leading and Trailing-Edge Perforations on the Thermohydraulic Performance of Rectangular-Winglet Vortex Generators

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In this study, the thermohydraulic performance of rectangular-winglet vortex generators (VGs) with different hole configurations (a central hole, dual-edge holes, a leading-edge hole, and a trailing-edge hole) was numerically investigated. In addition, airflows through solar air heater ducts enhanced with these VGs were simulated using the realizable k-epsilon model with a wall function at Reynolds numbers (Re) from 3000 to 20,000 and a Prandtl number (Pr) of 7.070. A constant blockage ratio, defined as the ratio of the VG surface area to the cross-sectional area of the airflow duct, was maintained. Overall, the results showed that the VG with a trailing-edge hole achieved the highest thermal enhancement factor (TEF), ranging from 1.394 to 1.600. The central-hole VG exhibited slightly lower TEF, between 1.424 and 1.578. In contrast, the dual-edge-hole and leading-edge-hole VGs produced lower TEFs, ranging from 1.301 to 1.541 and 1.282 to 1.567, respectively. Several temperature distribution plots, velocity plots, and pathline plots were utilized to investigate the cause of the performance differences. These visualizations ultimately revealed that the observed variations in thermohydraulic performance were attributed to distinct flow structures induced by the different hole configurations and Re values. In light of these findings, it can be concluded that the trailing-edge-hole VG is the most preferred option, while the central-hole VG serves as a viable alternative, offering nearly equivalent performance. In contrast, the leading-edge-hole and dual-edge-hole VGs are less efficient and not recommended.

Keywords: friction factor; hole position; leading edge; Nusselt number; perforated winglet vortex generator; solar air heater; trailing edge

Summary

- Hole placement on the leading or trailing edge of vortex generators (VGs) affects Nu and f differently, depending on Re .
- The trailing-edge-hole VG generally demonstrates the highest thermal enhancement factor (TEF).
- The central-hole VG exhibits TEF values nearly equivalent to those of the trailing-edge-hole VG.
- Performance differences are primarily due to distinct vortex structures induced by hole placement.

1. Introduction

Enhancing heat transfer performance is a fundamental objective in the design of solar air heaters. One effective approach to achieve this is through the incorporation of surface roughness geometries, which promote turbulence and enhance convective heat transfer between the air and the absorber surface. Oyakawa et al. [1] experimentally investigated the fluid flow and heat transfer characteristics of an endwall equipped with a 7×7 array of short rectangular plate fins inside a 50 mm high rectangular duct using an infrared imaging technique. The

rectangular fin arrays were arranged in coangular and zigzag patterns with varying pitch ratios. For Reynolds numbers (Re) ranging from 18,750 to 62,500, the results showed that increasing the distance between fins in the flow direction enhanced heat transfer in the coangular pattern, whereas decreasing this distance improved heat transfer in the zigzag pattern. Moreover, the zigzag pattern with a pitch ratio of 2 exhibited the highest heat transfer performance among all configurations tested. Islam et al. [2] conducted experiments to investigate the airflow reattachment positions of vortices formed behind short rectangular fins mounted on an endwall surface at an inclination angle of 20° . The results revealed that flow reattachment occurred at a distance of 40 mm (twice the fin length), and that the heat transfer rate was enhanced by a factor of 4 compared with that of a finless duct.

Once again, an endwall equipped with a 7×7 coangular-type array of short rectangular plate fins inside a channel was used in the study by Islam et al. [3] to measure its heat transfer characteristics. The objective was to investigate how the heat transfer performance was affected by the height ratio between the channel and the fin height. The resulting performance graph indicated that a channel-to-fin height ratio of 2 yielded the highest thermal performance within the tested range of 2–5. The authors concluded that at lower ratios, the separation space between the channel wall and the top surface of the fins became narrower, which accelerated the mainstream airflow velocity. This increase in velocity caused flow separation from the fin edges, leading to vortex formation that enhanced heat transfer. The use of solid vortex generators (VGs) was also demonstrated by Liang et al. [4], who presented methods that provide higher heat transfer performance with a comparatively lower pressure penalty in a circular tube equipped with winglet VGs. In their study, four winglet inclination angles (0° , 10° , 20° , and 30°), five winglet attack angles (0° , 10° , 20° , 30° , and 45°), and three winglet lengths (10, 15, and 20 mm) were investigated. They found that the Nusselt number and friction factor increased with increasing attack angle or winglet length but decreased with increasing inclination angle. These results also demonstrated that winglet VGs could generate longitudinal and transverse vortices, inducing both impingement flow and recirculation zones, which led to higher heat transfer while maintaining a comparatively lower pressure drop.

Some VGs used in heat exchanger ducts are perforated to enhance their thermohydraulic performance compared with their solid counterparts. These perforations are typically added as a single-hole positioned at or near the central region of a winglet VG. Skullong et al. [5] combined wavy grooves with centrally perforated delta-wing VGs on the absorber plate under a uniform heat flux to improve the thermal performance of a solar air heater channel. They achieved $\sim 37.7\%$ – 46.3% and 1.5% – 12.5% higher performance compared with that achieved using grooves alone and a combination of grooves with nonperforated VGs, respectively. They further [6] enhanced convective heat transfer in a heated circular tube using pairs of delta winglets with a central triangular perforation that were arranged inline on a perforated cross-shaped tape. These perforations reduced friction loss,

achieving a maximum thermal enhancement factor (TEF) of 1.902, higher than that achieved using nonperforated winglets (1.72).

In addition, Chamoli et al. [7] developed VGs with four central perforation levels (4%, 8%, 12%, and 16%) and three relative pitch lengths (2, 4, and 6). They found that the heat transfer and pressure drop increased with decreasing pitch lengths and perforation levels, whereas the TEF increased with increasing perforation levels. Promvong et al. [8] numerically investigated a centrally perforated delta-winglet mounted on a solar air heater absorber plate and found that its Nusselt number (Nu) and friction factor (f) were 17.1–78.21 and 3.92–5.9 times higher than a flat plate, respectively. The absorber plate thus achieved a peak performance factor of ~ 2.1 . A modified version of this winglet with a circular flap partially covering the hole, i.e., the flapped delta-winglet, achieved a further 2.9% improvement with a TEF of ~ 2.16 . The thermal enhancement and pressure drop in a circular tube equipped with radially arrayed winglet VGs mounted at different orientations were experimentally investigated by Liang et al. [9]. Several parameters were varied to examine their effects on the thermohydraulic characteristics of the tube, and one of these was the porosity ratio, which represented the degree of perforation in the VGs. The results showed that Nu increased as the porosity ratio rose from 0% to 10%, then decreased with further increases in the ratio. This behavior was attributed to the fact that moderate perforation levels enhance flow mixing through jets formed by the winglet holes, thereby strengthening convective heat transfer. However, as the porosity ratio approached 100%, the flow patterns became similar to those in a smooth tube, resulting in a decrease in Nu . Borrajo Pérez et al. [10] numerically examined centrally perforated delta winglets on channel surfaces by varying transverse spacing and flow configurations. They found that the effectiveness of VGs increased with increasing Re , achieving a performance evaluation criterion of 1.075 under upward flow with downstream-located holes and maximum transverse spacing. Huang and Liu [11] optimized delta-winglet parameters such as position, attack angle, and perforation radius for heat dissipation in a pin-fin heat sink. Using the CFD-ACE + solver and Levenberg–Marquardt algorithm, they showed that a higher Re favored larger perforations, attack angles, and winglet gaps; these factors enhanced TEF while reducing pressure loss.

Promvong et al. [12] studied louvered V-winglet VGs mounted on metal tapes in tubular heat exchangers. The relative pitch (0.5–1.5) and louver angle (10° – 90°) were varied while maintaining a fixed winglet height of 0.2 and attack angle of 30° . A maximum thermohydraulic performance factor of ~ 2.48 was achieved at a pitch of 1.0 and a louver angle of 30° . Wang et al. [13] experimentally and numerically analyzed the VG geometry under three height ratios, four central perforation levels, and three pitch ratios. Central perforations reduced the turbulent kinetic energy in the wake, thereby decreasing the local heat transfer and friction losses. However, pitch adjustment considerably improved the thermohydraulic performance, and Nu increased by 68%–257% relative to smooth tubes. They [14] also examined rectangular winglets

with central perforation levels of 5%, 10%, 20%, and 40% and pitch distances of 15, 30, and 45 mm. Higher perforation levels produced stronger jet flows that weakened downstream vortices, thereby reducing heat transfer and pressure losses. Expanding on this, Wang et al. [15] mounted punched curved rectangular winglets on the opposite walls of a microchannel. Under varying opening angles (0° – 15°) and pitch lengths (15, 30, and 45 mm), these generators induced mixed vortices that considerably enhanced the local heat transfer.

However, these perforations were placed eccentrically, away from the center, in some studies either to improve thermohydraulic performance or due to geometric constraints. Lu and Zhou [16] investigated the thermal and flow characteristics of both plane and curved VGs with and without perforations. In their study, holes were placed at the center and near the edges. They showed that perforated designs provided better heat transfer enhancement and lower flow resistance than solid designs, and optimal performance was achieved when the holes were positioned closer to the leading edge. Wang et al. [17] studied VGs mounted on opposite channel walls for different hole diameters (0.05, 0.15, and 0.25 mm) and pitch lengths (15, 30, and 45 mm). Larger holes generated stronger jet flows that suppressed longitudinal and corner vortices, and the largest holes eliminated some vortices entirely. In addition, Wang et al. [18] experimentally and numerically analyzed five hole positions on curved rectangular winglets in a microchannel heat exchanger. At $Re = 2500$, a perforation near the leading edge ($\theta = +12^\circ$) with a pitch of 15 mm yielded a TEF of 1.29.

The influence of both single and multiple perforations uniformly distributed on the surface of rectangular winglets on their thermohydraulic performance has been analyzed. For example, Hasan et al. [19] performed a numerical comparison of 12 winglet shapes with uniformly punched holes. The Gothic-type VG reduced outlet temperature by 12% more than the discrete ribbed generator and 4% more than a perforated louvered strip design. Heriyani et al. [20] studied concave rectangular-winglet pairs with uniform perforations in rectangular ducts containing six heated tubes. Their inline configuration achieved a lower cost-benefit ratio (3.56) compared with the staggered configuration, indicating better thermal performance relative to pressure drop. Similarly, Saini et al. [21] proposed solid and perforated curved trapezoidal winglets and tested them in fin-and-tube heat exchangers with four inline circular tubes. The configuration with six holes exhibited the best overall thermohydraulic performance. Saini et al. [22] further reported that punching six holes in a curved delta winglet VG yielded superior thermohydraulic performance compared with configurations containing zero, two, or three holes. In their study, both solid and perforated curved delta winglet VGs were installed within a fin-and-tube heat exchanger to numerically investigate their influence on the thermal-hydraulic performance factor (or TEF) and the performance evaluation criteria. In a subsequent study, Saini and Shah [23] modified curved delta winglet VGs by bending them into wavy shapes to enhance heat transfer performance. Their investigation examined the VGs with varying numbers of waves (ranging from one to seven) installed in a fin-and-

tube heat exchanger. The results demonstrated that the seven-wave curved delta winglet VG outperformed the other configurations. Moreover, the addition of circular perforations further enhanced the thermohydraulic performance of the heat exchanger. Lertnuwat [24] numerically investigated the impact of hole numbers and their placement on rectangular winglets in solar air heaters. The single-hole configuration achieved the highest average Nu , whereas the nine-hole configuration performed the worst. However, f remained relatively stable across configurations.

Promvong et al. [25] experimentally evaluated louver-perforated V-type baffle VGs in a constant heat-flux tube. Two array orientations, namely “V-down” (V-apex upstream) and “V-up” (V-apex downstream), were tested using a fixed attack angle (52°), relative height (0.3), and pitch (1.0). By varying six louver flap angles and three hole sizes and positions, the optimal setup was found to be a louver angle of 20° at the trailing end of the baffle. The V-down and V-up configurations achieved relative Nu values that were 5.9 and 6.38 times, respectively, higher than the solid baffles along with reduced friction. Similarly, Suchatawat et al. [26] combined experimental and numerical approaches to study louver-punched V-type winglets on solar receiver absorber plates. They showed that repositioning louver holes considerably enhanced the thermal performance of these plates. Wang et al. [27] introduced rectangular winglets with uniformly distributed perforations and analyzed their effects under turbulent conditions in a uniformly heated tube. Four angles of attack (0° , 15° , 30° , and 45°); four perforations (1, 2, 4, and 6); and three pitch lengths (25, 50, and 75 mm) were examined. Results showed that heat transfer improved with increasing number of holes and the angle of attack considerably influenced Nu and f .

In all the studies presented above, perforations were located between the leading and trailing edges of rectangular winglets, which is a standard design approach. However, recent studies have explored perforations placed specifically at the edges. Wang et al. [28] investigated rectangular winglets with holes located at vertical distances of 3, 5, and 7 mm and hole heights of 1, 3, and 6 mm. Lower-positioned holes generated stronger jet flows and improved thermal enhancement. Wang et al. [29] examined perforated delta winglets by varying the inclination angles (30° , 60° , and 90°); number of winglets (2, 4, and 6); and pitch ratios (4.89, 3.27, and 1.63). The highest TEF of 1.29 was achieved at a pitch ratio of 1.63, inclination angle of 30° , and number of winglets of 2 under an Re of 9090. Wang et al. [30] studied delta winglets under turbulent flows and found that larger hole areas reduced heat transfer and friction among holes with equal areas, whereas those with greater width and smaller height produced better thermal enhancement. Notably, Wang et al. [28–30] studied rectangular holes, which are less common than circular perforations in VGs.

Circular perforations were incorporated in the study conducted by Hassan et al. [31], who experimentally investigated the comparative performance of a flat-plate double-pass solar air heater employing solid and perforated V-corrugated absorbers. The perforated absorber contained

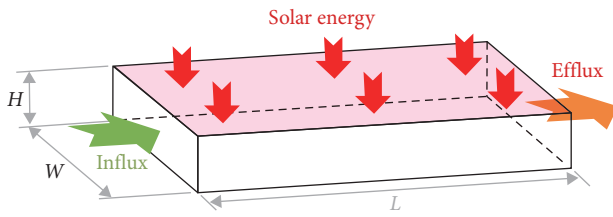


FIGURE 1: Schematic of the heat exchanger duct within a solar air heater.

an array of circular holes, each with a uniform diameter of 5 mm. These holes were evenly distributed across the absorber surface, with some positioned along the leading and trailing edges. Their results indicated that the perforated absorber consistently exhibited higher daily thermal efficiency than the solid absorber under all tested flow conditions. Another example of circular perforations on the trailing edge of a VG was reported by Khargotra et al. [32]; however, in this case, the working fluid was water instead of air. A solar water heating system integrated with delta-winglet VGs, each perforated with a circular hole on its trailing edge, was investigated to compare the friction factor, Nusselt number, and thermohydraulic performance under varying Reynolds numbers, angles of attack, and pitch ratios. Using a hybrid multicriteria decision-making approach that combined the analytic hierarchy process (AHP) and the additive ratio assessment (ARAS) method, the analysis indicated that the design configuration with an angle of attack of 45° , a Reynolds number of 1200, and a pitch ratio of 1 best satisfied the predefined performance criteria.

In summary, various surface roughness geometries have been employed to enhance the performance of heat exchangers. Some studies [1–4] have focused on simple solid winglet VGs, while others [5–27] have investigated perforated VGs. In these latter investigations, the perforations were typically located at or near the central region of the VGs, between the leading and trailing edges. These central-hole VGs have been commonly utilized in thermohydraulic performance studies, whereas edge-hole VGs have been less frequently explored. Only a limited number of studies have investigated edge-hole VGs [28–32]. However, these studies used edge-hole VGs solely to improve the thermohydraulic performance of heat exchangers without comparing their performance to that of central-hole VGs. Since the leading and trailing edges of winglet VGs critically influence flow development, and perforations placed specifically at these edges have not been sufficiently studied compared to those at the central region, the present study investigated the thermohydraulic performance of rectangular-winglet VGs with circular holes positioned at different locations, focusing on the leading and trailing edges. The obtained results were compared with those of a central hole.

2. Computational Setup

2.1. Heat Exchanger Duct and VG. Figure 1 shows the solar air heater comprising a heat exchanger with a rectangular duct. Based on the experimental parameters reported by

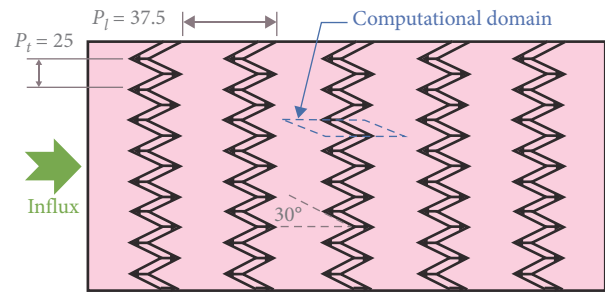


FIGURE 2: Schematic of the zigzag pattern for vortex generators on the absorber plate (dimensions in mm).

Skullong et al. [33], the dimensions of the heat exchanger were set to a width (W) of 200 mm, a height (H) of 25 mm, and a length (L) of 800 mm. Airflow passed through the duct from front to back. An absorber plate was positioned on top of the duct (highlighted in pink) to capture radiant solar energy and transfer it to the airflow, and the remaining duct walls were insulated.

Rectangular-winglet VGs were affixed to the inner surface of the absorber plate with a superglue. These VGs were arranged in a zigzag pattern (Figure 2), with an angle of attack of 30° relative to the incoming airflow. The longitudinal pitch (P_l) was 37.5 mm, and the transverse pitch (P_t) was 25 mm. Air entered the duct from the left and flowed to the right.

The rectangular-winglet VG had a height (e) of 12 mm and a base length (c) of 25 mm (Figure 3). Different hole configurations were integrated into the VGs, namely a central hole, dual-edge holes, a leading-edge hole, and a trailing-edge hole, to investigate their effects on the thermohydraulic performance of VGs. For the central-hole configuration (Figure 3a), a 5 mm diameter circular hole was created exactly at the center of the VG surface. In the dual-edge-hole configuration (Figure 3b), two semicircular holes of 5 mm diameter were created at the midpoints of both the leading and trailing edges, with the same perforation area as the central hole. For the leading-edge-hole configuration (Figure 3c), a single semicircular hole of 7.071 mm diameter was created at the leading edge to ensure that the perforation area was the same as that of the central-hole configuration. The trailing-edge-hole configuration (Figure 3d) is a mirror image of the leading-edge-hole configuration, with a semicircular hole of 7.071 mm diameter positioned at the trailing edge.

2.2. Computational Domain. The airflow rapidly becomes periodic after entering the heat exchanger duct and passing just a few winglet VGs, filling most of the duct. To save computational resources, the computational domain was designed as an oblique rectangular prism that enclosed a single-winglet VG and its immediate surroundings. Figure 2 shows the top view of this domain (indicated by blue dashed lines), and Figure 4 shows a pictorial representation of a domain for a VG with a leading-edge-hole configuration. The dimensions of the domain were set as follows: 12.5 mm width (half of the transverse pitch, P_t) and 37.5 mm length (the longitudinal pitch, P_l). These pitches corresponded to the zigzag patterns shown in Figure 2. The domain height (H) was

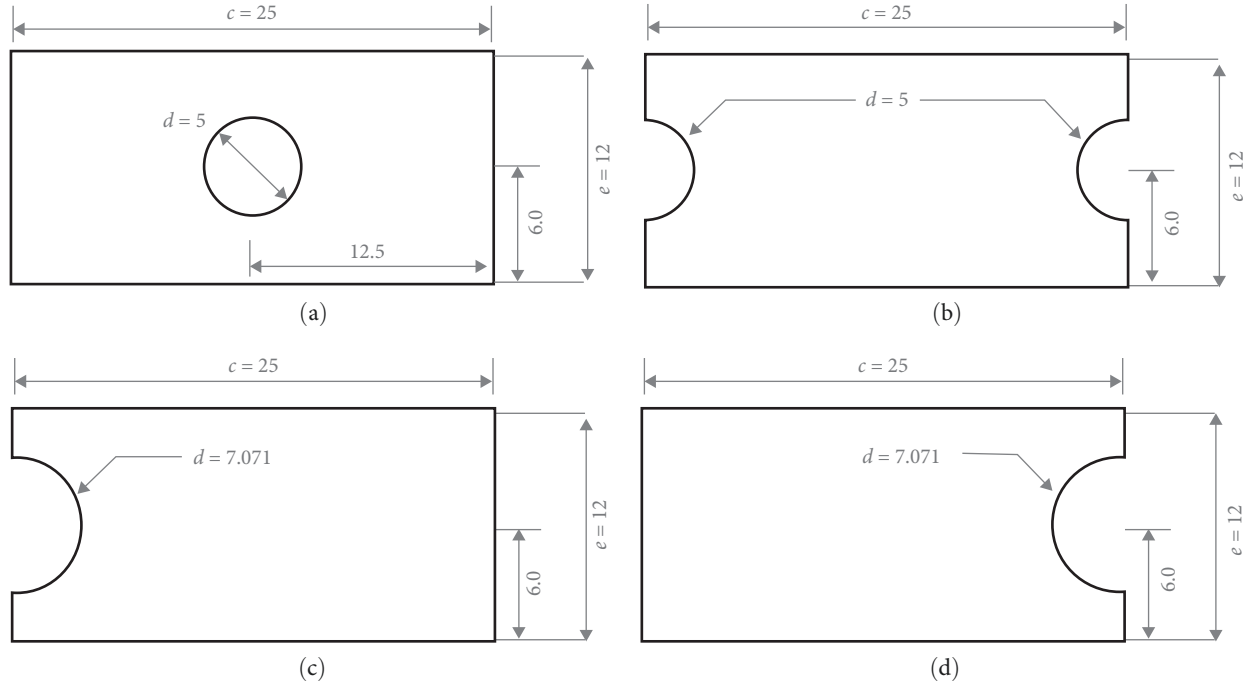


FIGURE 3: Schematic of the four hole configurations in rectangular-winglet vortex generators (dimensions in mm). (a) Central hole. (b) Dual-edge holes. (c) Leading-edge hole. (d) Trailing-edge hole.

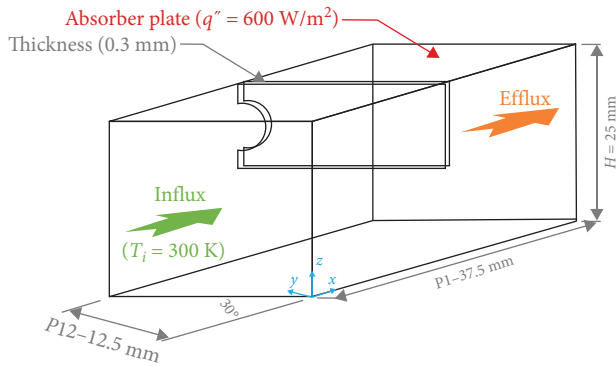


FIGURE 4: Pictorial representation of the computational domain, showing the space encompassing a winglet vortex generator with a leading-edge-hole configuration.

25 mm, matching the height of the heat exchanger duct. Its oblique angle was aligned with the angle of attack (30°), as shown in Figure 2, making the inlet and outlet planes parallel to the rectangular-winglet VG that was 0.3 mm thick. For the coordinate system, the x , y , and z axes were arranged with the z axis parallel to the height of the heat exchanger duct. The origin was placed at the lower-right corner of the inlet plane.

2.3. Boundary Conditions. The top surface, serving as the absorber plate, was treated as a no-slip wall. This surface released a constant heat flux (q'') of 600 W/m^2 to warm the airflow underneath it. Consistent with Skullong et al.'s study [33], the winglet VG was attached to the absorber plate with an insulating superglue to prevent heat transfer from the absorber plate to the winglet VG. Therefore, the insulated no-slip wall boundary condition was applied to all surfaces of

the winglet VG. The bottom surface of the computational domain was defined as the insulated no-slip wall, whereas the left and right surfaces were treated as symmetric planes.

Periodic inlet and outlet boundaries were imposed on the front and back sides of the computational domain, as shown in Figure 4. For a given Re , the mass flow rate on the inlet cross-sectional area was determined as follows:

$$\dot{m}_i = \rho \bar{u}_i P_t H / 2, \quad (1)$$

The area-averaged velocity ($\bar{u}_i$) at the inlet plane was calculated as follows:

$$\bar{u}_i = (\mu Re) / (\rho D). \quad (2)$$

Six specific Re values were chosen for simulations: 3000, 5000, 7000, 10,000, 15,000, and 20,000. The hydraulic diameter (D) was calculated as follows:

$$D = 2HW / (H + W). \quad (3)$$

The influx temperature (T_i) was held constant at 300 K.

2.4. Computational Model. Skullong et al. [33] conducted experiments and simulations on airflow in heat exchanger ducts equipped with various perforated VGs and found that the realizable k-epsilon turbulence model with standard wall functions outperformed the RNG k-epsilon and SST k-omega. Since the present simulations were validated against the empirical correlations proposed by Skullong et al. [33], the realizable k-epsilon model with standard wall

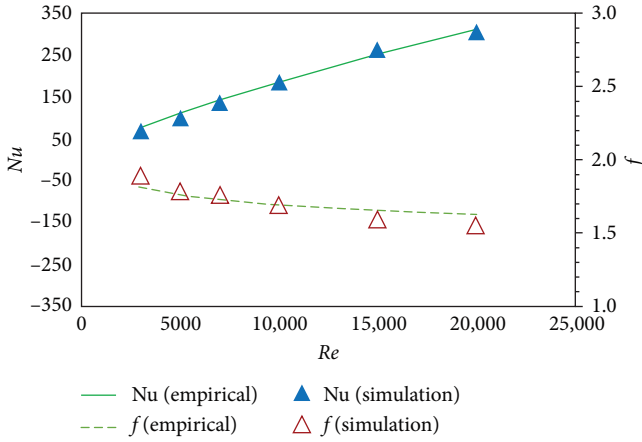


FIGURE 5: Validation of simulated results: a comparison between empirical and simulated Nusselt number (Nu) and friction factor (f) values, presented as functions of Reynolds number (Re).

functions was adopted for consistency and due to its high accuracy in modeling steady, incompressible, and turbulent flows.

Air was used as the working fluid, and its properties considered constant: density (ρ) of 1.177 kg/m^3 , dynamic viscosity (μ) of $1.846 \times 10^{-5} \text{ kg/m}\cdot\text{s}$, specific heat capacity (C_p) of $1004.9 \text{ J/kg}\cdot\text{K}$, and thermal conductivity (K) of $0.02624 \text{ W/m}\cdot\text{K}$. All simulations were conducted using ANSYS Fluent with the pressure-based solver. Additionally, the SIMPLE pressure-velocity coupling scheme and Rhie–Chow distance-based flux type were employed in the solution-methods setting.

2.5. Parameters of Interest. The thermohydraulic properties of solar air heaters are commonly assessed using two key parameters: Nu and f . Nu is defined as follows:

$$Nu = \bar{h}D/K, \quad (4)$$

where $\bar{h}$ is the area-averaged convective heat transfer coefficient represented as follows:

$$\bar{h} = q'' / (\bar{T}_{\text{sur}} - \bar{T}_{\text{air}}). \quad (5)$$

In this context, q'' is the 600-W/m^2 heat flux transferring from the absorber plate. $\bar{T}_{\text{sur}}$ is the area-averaged temperature of the absorber plate, and $\bar{T}_{\text{air}}$ is the mass-averaged temperature of air within the domain. f is calculated as follows:

$$f = \{(\Delta p/P_l)D\} / \left(\frac{1}{2}\rho\bar{u}_i^2\right), \quad (6)$$

where Δp is the area-averaged pressure drop between the inlet and outlet planes, P_l is the length of the longitudinal pitch, D is the hydraulic diameter, and $\bar{u}_i$ is the area-averaged velocity of the incoming air.

2.6. Verification of the Simulation Results. The simulated Nu and f were validated against the empirical correlations proposed by Skullong et al. [33] expressed as follows:

$$Nu = 0.194Re^{0.742}Pr^{0.4}(A_h/A_w)^{-0.0625} \text{Err.} \pm 4.7\%, \quad (7)$$

$$f = 0.1814Re^{-0.056}(A_h/A_w)^{-0.1634} \text{Err.} \pm 5.4\%, \quad (8)$$

where the Prandtl number (Pr) was set to a constant value of 0.7070. A_h and A_w , representing the total surface area of holes punched in the winglet VG and the surface area of an unperforated winglet VG, respectively, are defined as follows:

$$A_h = n(\pi d^2/4), \quad (9)$$

$$A_w = c \times e. \quad (10)$$

Figure 5 compares the simulation results with the empirical results calculated using Equations (7) and (8), revealing their consistent trends. The simulated Nu and f values closely match their empirical counterparts at the same Re . Thus, the simulation results aligned well with the empirical data and were therefore deemed acceptable.

2.7. Criteria for Solution Convergence and Grid Independence. Simulation convergence was determined based on residual values. These values were defined as the ratio of the absolute difference between the summed right-hand-side and left-hand-side terms of the discretized governing equations to the absolute sum of the right-hand-side terms. For a simulation to be considered converged, all residual values for the governing equations had to be less than 10^{-5} . Herein, the continuity, momentum, energy, turbulent kinetic energy, and turbulent dissipation rate equations were considered the governing equations.

Three mesh configurations were used to ensure grid independence, namely coarse, medium, and fine, comprising $\sim 38,300$, $48,200$, and $58,300$ elements, respectively. Grid independence was assessed by analyzing the relative changes in Nu and f . Figure 6 shows the percentage differences in Nu and f between the coarse, medium, and fine meshes for all four hole configurations. For every case, the difference in Nu and f between the medium and fine meshes was less than 1.00%.

The employed meshes were created using the default meshing method of ANSYS Fluent. The physical and solver preferences were set to “CFD” and “Fluent,” respectively. In addition, an advanced edge-sizing command with a dual bias (commonly referred to as a bell-curve bias) was applied. This command constrained the meshing process to generate smaller elements near the corners (including the regions adjacent to the top and bottom walls) and larger elements toward the center along the edges of the inlet and outlet planes, using a bias factor of 2.0.

Figure 7 graphically represents the grid independence for Nu and f at $Re = 10,000$. All the curves converge to an asymptote as the number of elements increases, indicating finer meshes. $\%_{\text{diff}Nu}$ and $\%_{\text{diff}f}$ were calculated specifically for the

Re	Elements	Nu	f	%diff $_{Nu}$	%diff $_f$
3000	37,880	67.6	1.873		
	47,122	67.7	1.898	0.12	1.32
	57,187	67.6	1.886	0.20	0.63
5000	37,880	99.1	1.818		
	47,827	99.0	1.790	0.16	1.52
	57,934	98.8	1.779	0.14	0.65
7000	37,880	136.6	1.786		
	47,427	135.6	1.761	0.67	1.44
	57,721	135.7	1.757	0.04	0.23
10,000	37,880	186.0	1.705		
	47,292	184.7	1.686	0.70	1.08
	57,411	184.4	1.681	0.16	0.28
15,000	37,880	264.0	1.606		
	47,387	261.4	1.590	0.99	0.98
	57,452	261.4	1.589	0.01	0.06
20,000	37,880	303.3	1.540		
	47,833	304.6	1.545	0.43	0.32
	57,630	305.1	1.549	0.16	0.23

(a)

Re	Elements	Nu	f	%diff $_{Nu}$	%diff $_f$
3000	38,557	68.3	2.050		
	48,799	68.5	2.020	0.20	1.49
	58,472	68.5	2.006	0.01	0.69
5000	38,557	100.0	1.999		
	48,149	99.6	1.966	0.36	1.64
	58,686	100.0	1.958	0.33	0.41
7000	38,557	138.1	1.965		
	48,132	136.7	1.939	0.97	1.31
	58,290	137.0	1.935	0.20	0.20
10,000	38,557	187.0	1.886		
	48,511	184.8	1.868	1.15	0.98
	58,497	184.7	1.864	0.08	0.19
15,000	38,557	263.0	1.785		
	48,166	258.9	1.768	1.55	0.93
	58,783	257.6	1.763	0.49	0.29
20,000	38,557	282.1	1.686		
	48,695	282.6	1.693	0.16	0.41
	58,769	283.0	1.695	0.16	0.11

(c)

Re	Elements	Nu	f	%diff $_{Nu}$	%diff $_f$
3000	38,548	67.5	1.994		
	48,016	67.1	1.978	0.54	0.80
	58,315	66.9	1.965	0.35	0.67
5000	38,548	99.3	1.939		
	48,235	98.1	1.909	1.22	1.57
	58,580	97.7	1.895	0.40	0.72
7000	38,548	136.7	1.905		
	48,526	134.8	1.875	1.41	1.55
	58,713	134.4	1.866	0.32	0.50
10,000	38,548	186.1	1.821		
	48,904	183.8	1.796	1.20	1.42
	58,575	182.9	1.788	0.49	0.40
15,000	38,548	263.2	1.719		
	48,121	260.5	1.698	1.02	1.19
	58,883	260.1	1.697	0.15	0.10
20,000	38,548	283.6	1.596		
	48,795	281.6	1.591	0.69	0.28
	58,412	281.2	1.592	0.15	0.05

(b)

Re	Elements	Nu	f	%diff $_{Nu}$	%diff $_f$
3000	38,178	69.0	1.929		
	48,798	68.8	1.959	0.25	1.56
	58,229	68.5	1.943	0.45	0.80
5000	38,178	102.1	1.885		
	48,963	100.6	1.852	1.53	1.74
	58,833	100.2	1.841	0.41	0.57
7000	38,178	141.0	1.851		
	48,406	139.2	1.817	1.27	1.84
	58,868	138.6	1.807	0.46	0.55
10,000	38,178	192.3	1.762		
	48,255	190.2	1.734	1.12	1.57
	58,862	189.6	1.729	0.29	0.30
15,000	38,178	273.7	1.652		
	48,516	271.1	1.628	0.96	1.45
	58,816	269.9	1.621	0.42	0.38
20,000	38,178	298.7	1.494		
	48,753	295.3	1.482	1.12	0.79
	58,418	294.3	1.483	0.34	0.03

(d)

FIGURE 6: Number of elements in computational domains and corresponding percentage differences in Nu and f for various hole configurations. (a) Central-hole configuration. (b) Dual-edge-hole configuration. (c) Leading-edge-hole configuration. (d) Trailing-edge-hole configuration.

central-hole configuration. Results for other Re values showed similar trends and were therefore omitted for brevity.

Based on the data shown in Figures 6 and 7, the simulation results obtained from the fine meshes were considered grid-independent and consequently selected for the analysis and discussion of results.

3. Results

The thermohydraulic performance of rectangular-winglet VGs with different hole configurations was compared with that with a central-hole configuration. To this end, Nu and f for the finest meshes of all edge-hole configurations in

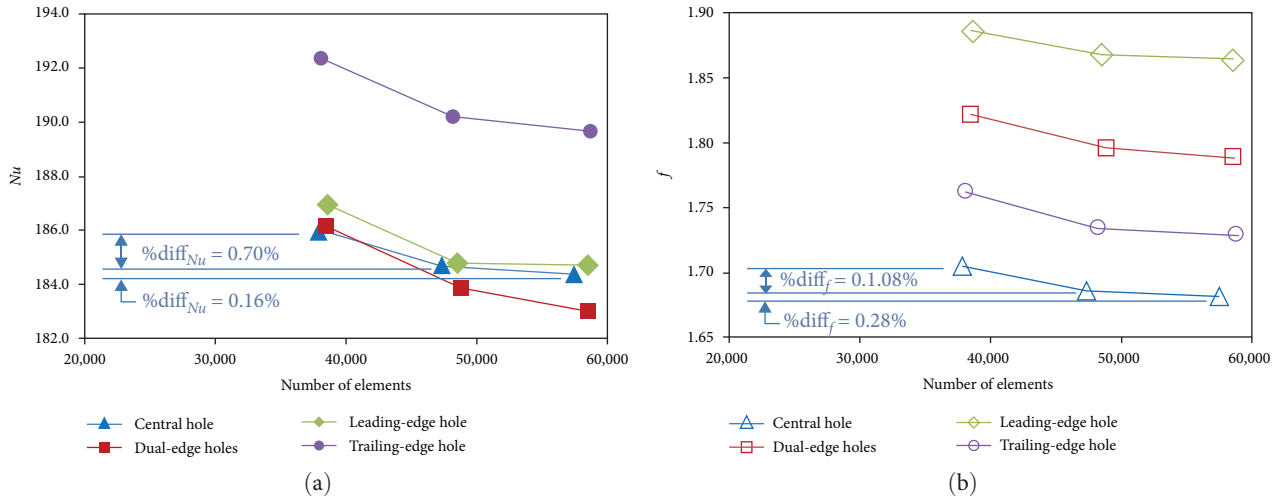


FIGURE 7: Examples of grid independence for Nusselt number (Nu) and friction factor (f) for the central-hole configuration at $Re = 10,000$. (a) Nu as a function of the number of elements (b) f as a function of the number of elements.

Re	$Nu/Nu_{\text{centralhole}}$		
	Dual-edge-hole	Leading-edge hole	Trailing-edge-hole
3000	0.990	1.014	1.014
5000	0.988	1.012	1.014
7000	0.990	1.010	1.021
10,000	0.992	1.002	1.028
15,000	0.995	0.985	1.033
20,000	0.922	0.928	0.965

(a)

Re	$f/f_{\text{centralhole}}$		
	Dual-edge-hole	Leading-edge hole	Trailing-edge-hole
3000	1.041	1.063	1.030
5000	1.065	1.101	1.035
7000	1.062	1.101	1.029
10,000	1.064	1.109	1.028
15,000	1.068	1.109	1.020
20,000	1.028	1.094	0.957

(b)

FIGURE 8: Normalized Nu and f relative to the corresponding values of the central-hole configuration for various edge-hole configurations. (a) Normalized Nu ($Nu/Nu_{\text{Central hole}}$). (b) Normalized f ($f/f_{\text{Central hole}}$).

Figure 6 (indicated in red) were normalized by the corresponding values of the central-hole configuration (indicated by blue numbers). Figure 8 shows these normalized Nu and f ranging from 0.922 to 1.047 and 0.957 to 1.109, respectively. The simulation scheme was validated using Equations (7) and (8), with $\pm 4.7\%$ and $\pm 5.4\%$ uncertainties, respectively; therefore, these uncertainties served as criteria for assessing the significance of the effect when holes were moved to the edges of the VGs.

Figure 8a shows that only two normalized Nu values (marked in red) are significantly different from unity, deviating by much more than the $\pm 4.7\%$ uncertainty criterion. These values correspond to the dual-edge-hole and leading-edge-hole configurations at $Re = 20,000$.

Figure 8b shows the additional results for the normalized f . For the dual-edge-hole configuration, when $5000 \leq Re \leq 15,000$, the normalized f values (marked in blue) slightly deviate from the $\pm 5.4\%$ uncertainty criterion. Conversely, for the leading-edge-hole configuration, all normalized f values (marked in red) significantly deviate from this criterion, except for the value at $Re = 3000$ (marked in blue). In contrast, the normalized f values for the trailing-edge-hole

configuration are insignificantly different from unity, indicating that this configuration offers similar thermohydraulic performance as the central-hole configuration.

Figure 9 visually clarifies the comparisons shown in Figure 8. Figure 9a specifically illustrates how the normalized Nu scatters around unity for all three hole configurations. The blue dashed lines represent the upper and lower bounds, defined by the $\pm 4.7\%$ uncertainty criterion. Notably, two data points at $Re = 20,000$, corresponding to the dual-edge- and leading-edge-hole configurations, fall significantly below the lower bound.

Figure 9b shows the scatter of normalized f . Similar to Figure 9a, blue dashed lines indicate the upper and lower bounds, but for this plot, they correspond to the $\pm 5.4\%$ uncertainty criterion. Nearly all data points for the leading-edge-hole configuration appear far above the upper bound, except for the one at $Re = 3000$, which only slightly falls above it. In comparison, only four data points for the dual-edge-hole configuration, specifically between $Re = 5000$ and $15,000$, fall slightly above the upper bound. In contrast, all the data points for the trailing-edge-hole configuration exist entirely within the area between the upper and lower bounds.

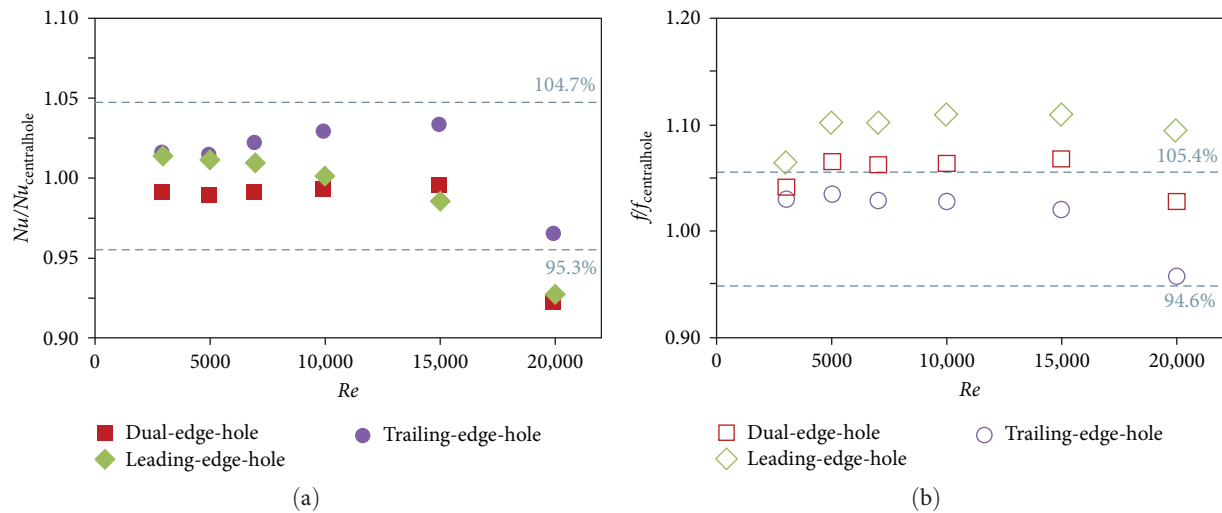


FIGURE 9: Visual comparison of (a) normalized Nusselt number ($Nu/Nu_{\text{Central hole}}$) and (b) normalized friction factor ($f/f_{\text{Central hole}}$) with their uncertainty criteria for various edge-hole configurations.

4. Discussion

Based on Figure 9, the results were evidentially categorized into two groups based on Re . For the first group containing data points at lower Re ($Re \leq 15,000$), the figure shows that f is significantly impacted. This impact is severe for the leading-edge-hole configuration and mild for the dual-edge-hole configuration. Conversely, Nu for all edge-hole configurations shows insignificant changes under these conditions.

For the second group containing data points at $Re = 20,000$, Nu for the dual-edge-hole and leading-edge-hole configurations is evidently affected. Concurrently, only the f of the leading-edge-hole configuration considerably changed from that of the central-hole configuration.

These changes are discussed subsequently using the coordinate system, selected views, and selected planes, which are illustrated in Figure 10 in a three-dimensional view. The origin of the coordinate system is located at the lower-right corner of the inlet plane. The x -direction runs along the longitudinal pitch, the z -direction points upward, and thus, the y -direction is oriented to the left. The blue, red, and yellow arrows represent the top, front, and side views, respectively. The orange plane indicates the zx plane at $y = 2$ mm.

4.1. Group 1: Results at the Lower Re ($Re \leq 15,000$). Due to their similarity and for brevity, the results at $Re = 10,000$ were selected for detailed discussion. The friction factors of the heat-exchanger ducts with different VGs were contributed by two main parts: the skin friction drag on the no-slip walls and the form drag over the VGs, with the latter being the dominant one. Consequently, the pressure distributions on the windward and leeward surfaces of the VGs were investigated as presented in Figure 11.

A notable similarity is observed across all cases regarding the pressure distributions on the windward surfaces. Specifically, the lower-edge area and the hole-perimeter area of each windward surface exhibit lower pressures compared to those of the majority of the surface. While subtle differences exist

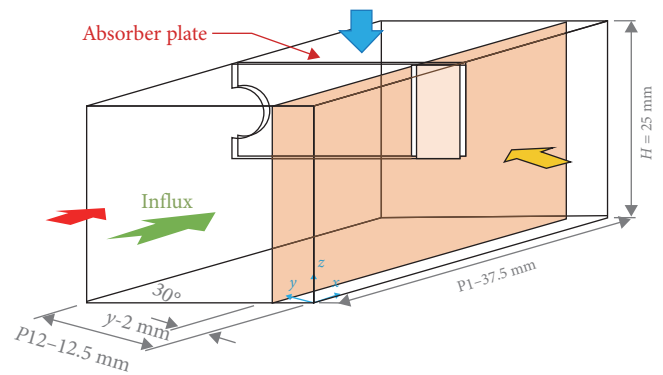


FIGURE 10: Coordinate system, with views and planes.

between cases, these variations are not substantial. Conversely, the pressure distributions on the leeward surfaces show a pronounced difference. A distinct very-low-pressure area is present underneath the leading-edge hole for the dual-edge-hole and leading-edge-hole configurations. This area is entirely absent in the other two cases. This difference thus caused a higher pressure difference and a higher f for the dual-edge-hole and leading-edge-hole configurations; these findings are consistent with the results obtained at $Re = 10,000$, as shown in Figure 9b.

The formation of those low-pressure areas on the leeward surface of VGs with dual-edge holes and leading-edge hole was investigated by examining their velocity fields. Figure 12 presents three-dimensional pictorial views of the four cases. The airflow within each case can be broadly categorized into three distinct regions based on altitude: lower, middle, and upper.

The airflows in the lower and middle regions are similar across all cases. In the lower region, each case is characterized by high-speed airflow moving approximately parallel to the x axis. This airflow primarily convects the heat toward the exit. In the middle region, the airflow moves toward the lower edge

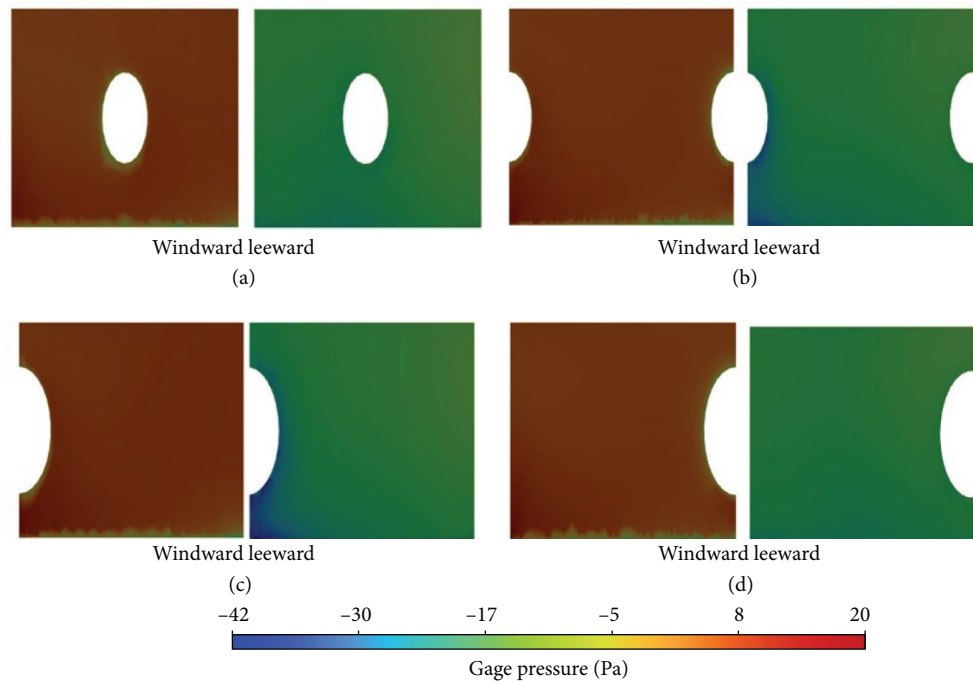


FIGURE 11: Front views of pressure distributions on the windward and leeward surfaces of the vortex generators at $Re = 10,000$. (a) Central-hole. (b) Dual-edge-hole. (c) Leading-edge-hole. (d) Trailing-edge-hole.

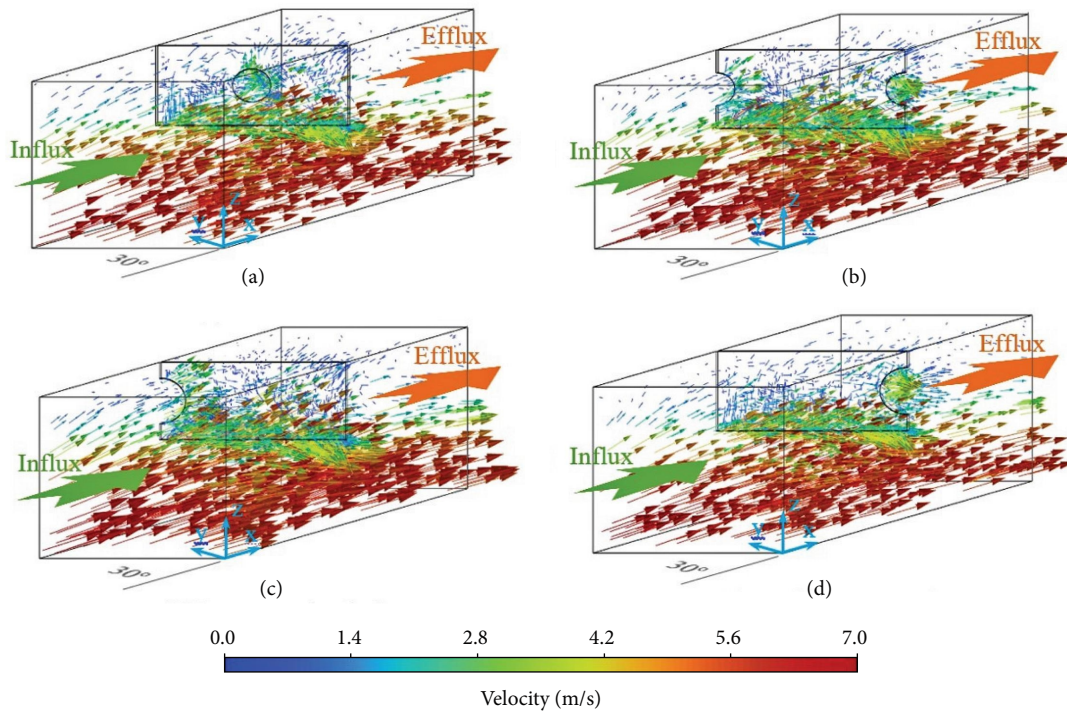


FIGURE 12: Three-dimensional pictorial views of the velocity plots for the four hole configurations at $Re = 10,000$. (a) Central-hole. (b) Dual-edge-hole. (c) Leading-edge-hole. (d) Trailing-edge-hole.

of the VG and flows beneath it, with most of the air passing along the right side of the VG, close to the trailing edge. Subsequently, the airflow rises and reverses direction. This portion of the airflow primarily forms a main longitudinal

vortex, which convects the heat from the absorber plate above down to the middle and lower regions of the airflow.

On the other hand, the airflows in the upper regions differ considerably due to varying hole positions. As these

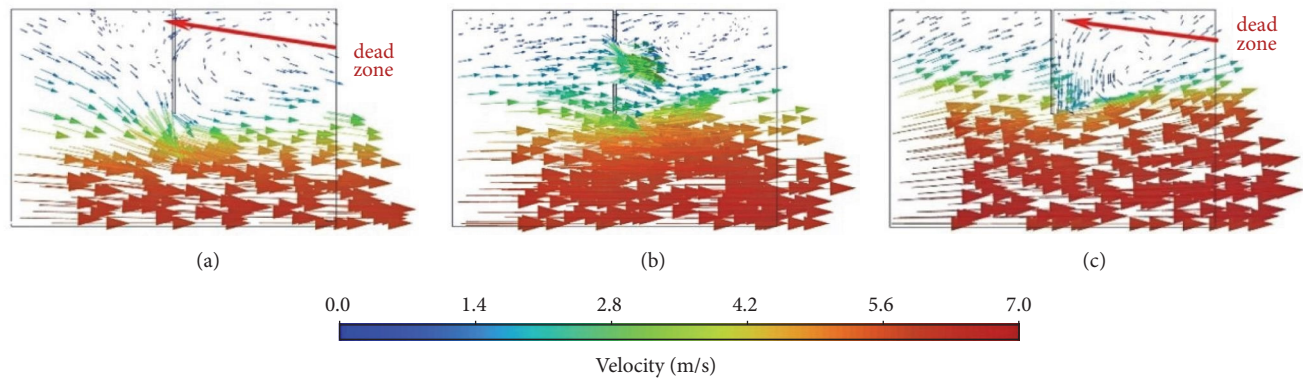


FIGURE 13: Side views of two-dimensional velocity plots on selected zx planes for the central-hole configuration at $Re = 10,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

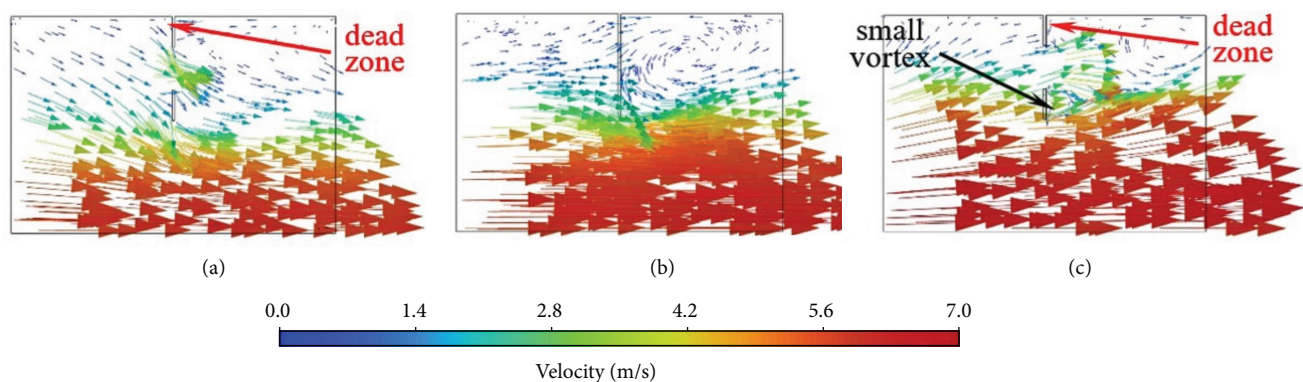


FIGURE 14: Side views of two-dimensional velocity plots on selected zx planes for the dual-edge-hole configuration at $Re = 10,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

three-dimensional pictorial views are quite complex and therefore ambiguous for observing detailed flow structures behind the VGs, side views of two-dimensional velocity plots on selected zx planes are shown in Figures 13–16.

Figure 13 shows the two-dimensional velocity plots on zx planes for the central-hole configuration at three distinct y locations: 0.00 mm (trailing edge), 6.25 mm (midspan of the VG), and 12.50 mm (leading edge). These plots reveal a large counterclockwise vortex on the leeward side of the VG, which is notably strong behind the leading edge (Figure 13c). A dead zone, where the airflow is nearly stagnant, is also observed in the corner between the VG's leeward side and the inner surface of the absorber plate, indicated by the red arrow. At midspan (Figure 13b), the central hole allows airflow to penetrate directly to the leeward side of the VG as an air jet, which reduces the vortex size by pushing it upward and eliminating the dead zone. Additionally, the vortex strength weakens significantly. The strength of the vortex behind the trailing edge (Figure 13a) is moderate, but its size occupies a larger area compared to those shown in Figure 13b,c. Another dead zone is observed in the corner between the VG's windward side and the inner surface of the absorber plate, as indicated by the red arrow.

Figure 14 shows two-dimensional velocity plots for the dual-edge-hole configuration on the same three zx planes

presented in Figure 13. Similar to the central-hole configuration, a large counterclockwise vortex is observed on the leeward side of the VG, along with two dead zones. The first dead zone appears in the corner between the VG's windward side and the inner surface of the absorber plate, as indicated by the red arrow in Figure 14a. Compared to Figure 13a, this dead zone is smaller due to the presence of the trailing-edge hole, which allows upper-region airflow to shortcut to the leeward side of the VG. As a result, the upper-region airflow on the windward side no longer needs to circumvent the bottom edge of the VG, reducing the turning radius required to avoid impacting the VG and thereby resulting in a smaller dead zone. Additionally, the vortex on the VG's leeward side is smaller than that shown in Figure 13a because the trailing-edge hole produces an air jet that pushes the vortex upward, an identical mechanism to that shown in Figure 13b. At midspan (Figure 14b), the vortex is larger and stronger than that shown in Figure 13b due to the absence of punched holes. This creates a large suction region behind the VG. However, no dead zone is observed in this plane; instead, the second dead zone appears at the leading-edge plane, indicated by the red arrow in Figure 14c. In this figure, the vortex is much smaller than that shown in Figure 13c because the air jet from the leading-edge hole pushes the vortex upward. Despite the air jet, the size of the dead zone is comparable to that in

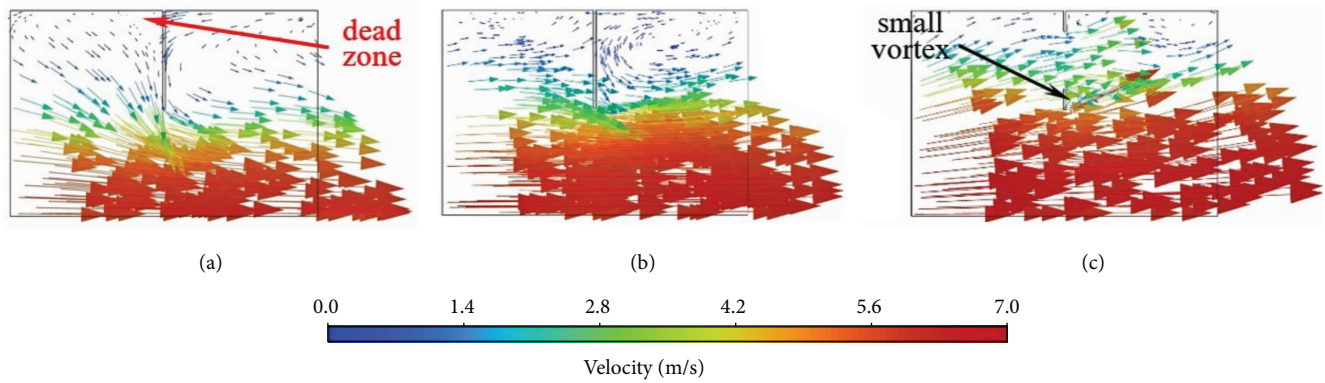


FIGURE 15: Side views of two-dimensional velocity plots on selected zx planes for the leading-edge-hole configuration at $Re = 10,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

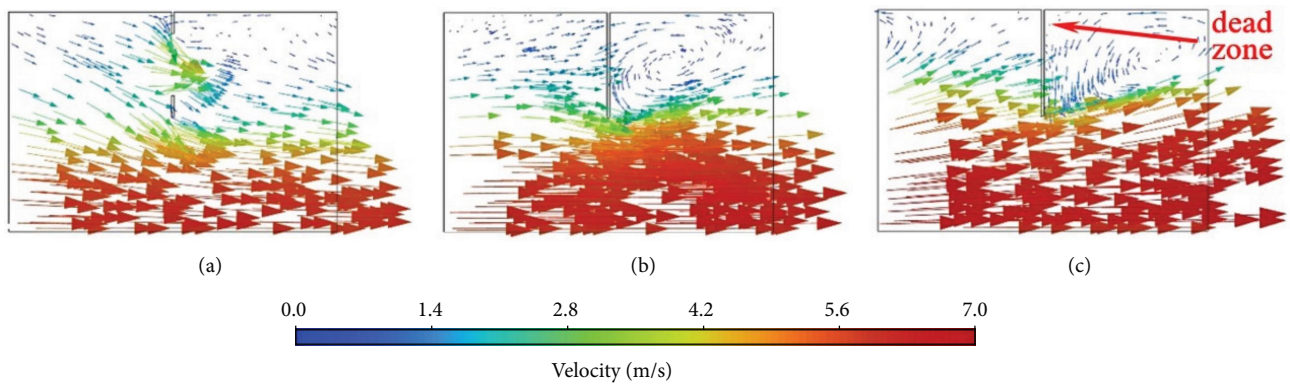


FIGURE 16: Side views of two-dimensional velocity plots on selected zx planes for the trailing-edge-hole configuration at $Re = 10,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

Figure 13c. This is likely because the impenetrable portion of the VG above the hole is large enough to accommodate the size of the dead zone.

Similar to Figure 13, the three selected zx planes are shown in Figure 15 to present the velocity plots for the leading-edge-hole configuration. Only one dead zone is observed in Figure 15a, and it is larger than that in Figure 13a. This is because the punched hole in the central-hole configuration is located closer to the dead zone. As a result, the airflow reaches the leeward side of the VG more quickly, reducing both the turning radius of the airflow pathline and the size of the dead zone compared to the leading-edge-hole configuration. However, the vortices behind the VG in Figures 13a and 15a are similar in both size and strength. At midspan (Figure 15b), as expected, the vortex is larger and stronger than that in Figure 13b due to the absence of the punched hole. Similarly, on the leading-edge plane (Figure 15c), the presence of the punched hole generates an air jet, which reduces the size of the vortex. This is why the vortex in Figure 15c is notably smaller than those in both Figures 13c and 14c. It is also noteworthy that the dead zone above the punched hole, as observed in Figure 14c, is not visible in Figure 15c. This is due to the smaller impenetrable portion of the VG above the punched hole. When this

portion is too small, it cannot support the formation of a dead zone behind it, as seen in Figure 15c.

For the trailing-edge-hole configuration, Figure 16 presents two-dimensional velocity plots on the same three zx planes shown in Figure 13. In Figure 16a, the presence of a punched hole results in no dead zone on the windward side of the VG, which is notably different from the other configurations. This observation suggests that the size of the punched hole plays a significant role in the absence of the dead zone. Specifically, when the punched hole is large, the impenetrable portion of the VG above the hole becomes too small to support the formation of a dead zone in front of it. In Figure 16b, the vortex behind the VG resembles those in Figures 14b and 15b, as no punched hole is present, which differs from the case in Figure 13b. In Figure 16c, a dead zone is observed, and it is larger than the one in Figure 13c, although the leading-edge hole is similarly absent in both figures. This is likely because the air jet in Figure 13b supplies some amount of airflow into the dead zone in Figure 13c, thereby suppressing its size. In contrast, the dead zone in Figure 16c does not receive this replenishment.

Although the velocity plots in Figures 13–16 are similarly characterized by large counterclockwise vortices and dead zones, a small but noticeable flow structure appears only in

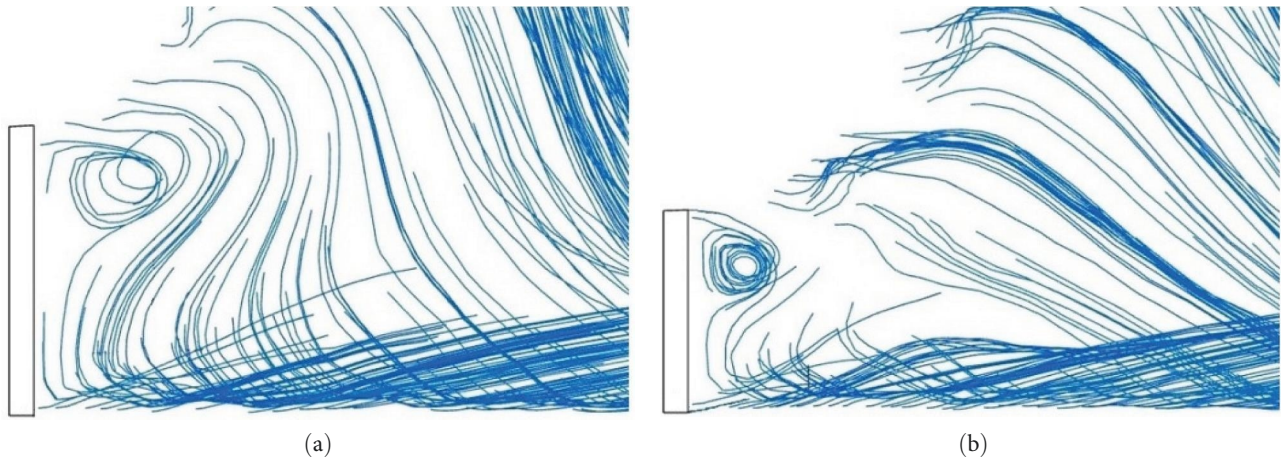


FIGURE 17: Side views of pathlines on an xz plane illustrating the small vortex structure on the leeward side for the (a) dual-edge-hole (at $y = 12.50$ mm) and (b) leading-edge-hole (at $y = 12.50$ mm) configurations at $Re = 10,000$.

the dual-edge-hole and leading-edge-hole configurations—namely, the small vortices indicated by the black arrow in Figures 14c and 15c. As these two small vortices are indistinct in the velocity plots, pathline plots are shown in Figure 17a,b for enhanced visibility of the results.

It is evident that, in Figures 14c and 15c, these vortices are induced by the air jets from the leading-edge holes and the airflows circumventing the bottom edges of the VGs. Clear evidence for these is provided by the flow fields on the leeward sides of the VGs in Figures 13c and 16c, where no small vortices are present. In these configurations, the main vortices are large enough to occupy nearly all of the spaces behind the VGs, preventing secondary vortex formation. However, in the dual-edge-hole and leading-edge-hole configurations (Figures 14c and 15c), the main vortices are pushed upward, leaving some spaces between the air jets and the bottom edges of the VGs. Since airflows barely reach these spaces, low-pressure zones are induced due to the suction effect. Subsequently, smaller vortices form within these spaces. This is analogous to the flow phenomenon where vortices emerge in the wake behind flow separation of a blunt body.

These small vortices within the flow fields of the dual-edge-hole and leading-edge-hole configurations provide evidence of the low-pressure areas forming on the leeward surfaces below the holes on the VGs (Figure 11b,c).

Regarding Nu at $Re = 10,000$ in Figure 9a, since $\bar{T}_{air}$ was almost constant for a fixed Re , and as per Equations (4) and (5), Nu exhibited an inverse relationship with $\bar{T}_{sur}$, that is, Nu increased with decreasing $\bar{T}_{sur}$. Figure 18 shows the temperature distributions on the inner surfaces of the absorber from a top view for the four cases, with the corresponding $\bar{T}_{sur}$ values indicated in the sub-figure captions.

Figure 18 shows the temperature distributions, revealing regions on the inner surfaces of the absorber with extremely high temperatures; this indicates that these areas have very low convection heat transfer and that the dead zones of the adjacent flow fields are below the inner surfaces of the absorber. As expected, in the presence of the central hole (Figure 18a), the dead zones present in the leeward leading edge and windward trailing edge of the VG. The temperature in the region of the

leeward leading edge is obviously much higher. In the presence of dual-edge holes (Figure 18b), the maximum temperature magnitude within the dead zones behind the leeward leading edge is reduced, but the area of the high-temperature region increases. As $\bar{T}_{sur}$ for this configuration is higher than that for the central-hole configuration, the overall impact of dead zones in this configuration increases.

When the hole is relocated to the leading edge (Figure 18c), it eliminates the peak temperature within the dead zones behind the leeward leading edge because no red spots are present in this region. However, the temperature within the dead zone in front of the windward trailing edge considerably increases. The distribution for this configuration is a mirror reflection of the central-hole configuration but with only one dead zone. This leads to slightly less overall impact from the dead zones, as indicated by the lower $\bar{T}_{sur}$ value for this configuration compared with that for the central-hole configuration (Figure 18a).

Figure 18d shows the temperature distribution when the hole is on the trailing edge. The peak temperature disappears from the windward trailing edge; however, the maximum temperature magnitude behind the leeward leading edge is higher than those in the other cases. This distribution is a mirror reflection of that observed for the leading-edge-hole configuration (Figure 18c), with an obviously higher peak temperature but a narrower hot area (green to red painted areas). However, based on the comparison of the $\bar{T}_{sur}$ values, the impact of the dead zones in this case is the lowest among all four configurations, with the $\bar{T}_{sur}$ values ranking from highest to lowest as follows: dual-edge-hole, central-hole, leading-edge-hole, and trailing-edge-hole.

Note that $\bar{T}_{sur}$ ranges from 305.92 to 306.11 K, and the central-hole configuration exhibits an intermediate $\bar{T}_{sur}$ (the second highest temperature). Consequently, the differences in $\bar{T}_{sur}$ relative to the central-hole configuration are 0.04 K for the dual-edge hole, -0.04 K for the leading-edge hole, and -0.15 K for the trailing-edge hole. These findings are consistent with those shown in Figure 9a. However, these differences are very small and only slightly affect Nu at $Re = 10,000$.

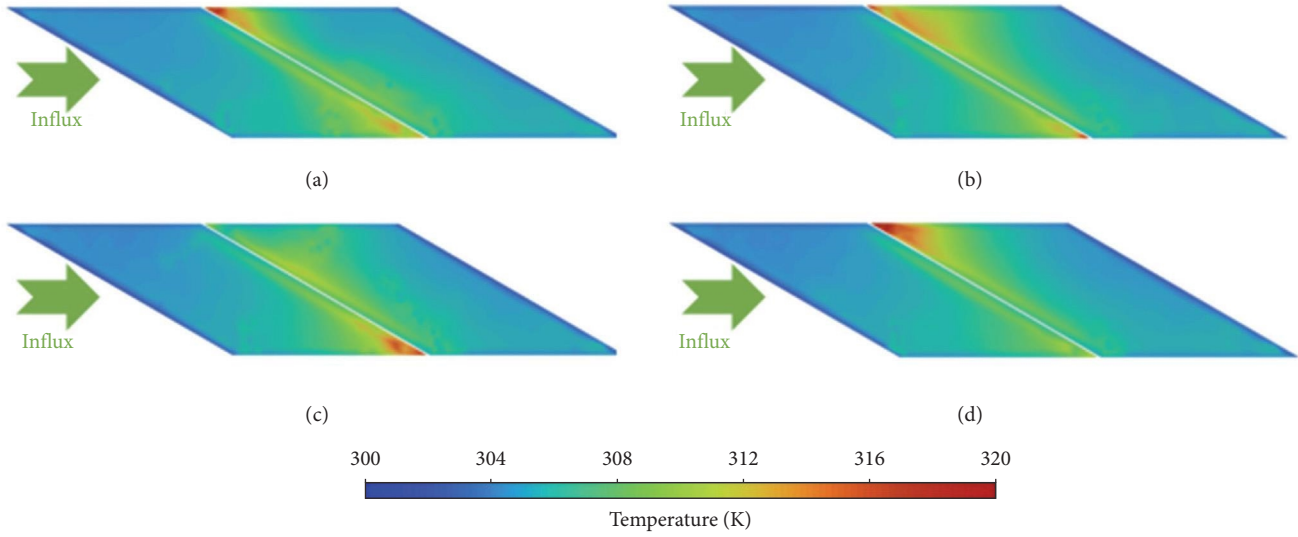


FIGURE 18: Top views of temperature distributions on the inner surfaces of the absorber for $Re=10,000$, with corresponding $\bar{T}_{sur}$ values. (a) Central-hole ($\bar{T}_{sur} = 306.07$ K). (b) Dual-edge-hole ($\bar{T}_{sur} = 306.11$ K). (c) Leading-edge-hole ($\bar{T}_{sur} = 306.03$ K). (d) Trailing-edge-hole ($\bar{T}_{sur} = 305.92$ K).

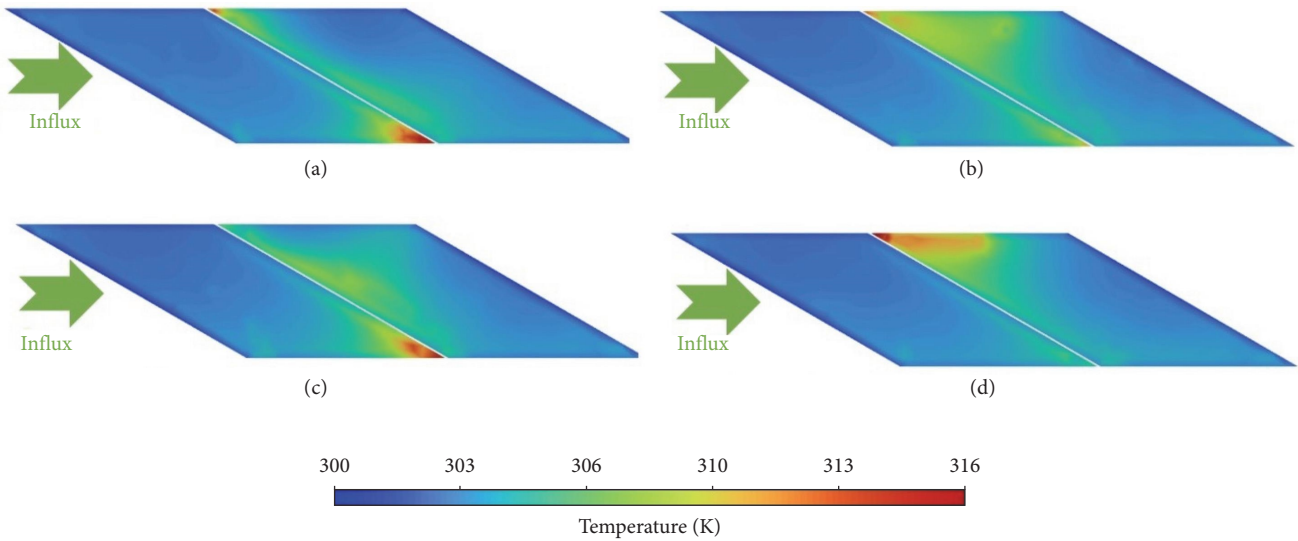


FIGURE 19: Top views of temperature distributions on the inner surfaces of the absorber for $Re=20,000$ with corresponding $\bar{T}_{sur}$ values. (a) Central-hole ($\bar{T}_{sur} = 303.63$ K). (b) Dual-edge-hole ($\bar{T}_{sur} = 303.94$ K). (c) Leading-edge-hole ($\bar{T}_{sur} = 303.87$ K). (d) Trailing-edge-hole ($\bar{T}_{sur} = 303.79$ K).

4.2. Group 2: Results Obtained at Higher Re ($Re = 20,000$).

Figure 9a shows that the normalized Nu values of all three hole configurations at $Re = 20,000$ are lower than unity, which is further explained using the top views of temperature distributions on the inner surfaces of the absorber. Figure 19a–d shows that the high-temperature regions emerging from the dead zones relate directly to the positions of holes for all four hole configurations. These figures briefly resemble Figure 18a–d; however, a noticeable difference is observed for the central-hole configuration, whereas the other cases show only subtle nuances. A comparison of Figures 19a and 18a show two hot areas but with their hottest spots located at different positions.

Specifically, the hottest spot is located on the windward trailing-edge in Figure 19a but on the leeward leading-edge in Figure 18a. This implies a substantial change in the flow structures for the central-hole configuration when the Re increases to 20,000.

In addition to comparing the hottest spots' locations, Figure 20 reveals an interesting point regarding the orders of $\bar{T}_{sur}$. For the central-hole configuration (highlighted in red), the order of $\bar{T}_{sur}$ changes from the second hottest temperature at $Re = 10,000$ to the coolest temperature at $Re = 20,000$. The orders of the other configurations, however, are the same for both Re conditions. At $Re = 20,000$, the $\bar{T}_{sur}$ of the central-hole

Re	Configurations $\bar{T}_{sur}$ in Kelvin			
	Hottest	Second hottest	Third hottest	Cooltest
10,000	Dual-edge (306.11)	Central (306.07)	Leading-edge (306.03)	Trailing-edge (305.92)
20,000	Dual-edge (303.94)	Leading-edge (303.87)	Trailing-edge (303.79)	Central (303.63)

FIGURE 20: Comparison of $\bar{T}_{sur}$ orders across different hole configurations, ranked from hottest to coolest, at $Re = 10,000$ and $Re = 20,000$.

configuration is the lowest; therefore, it differs from those of the dual-edge-hole and leading-edge-hole configurations more significantly than it does at $Re = 10,000$. This explains why the data points of the normalized Nu for these two configurations decrease severely in Figure 9a at $Re = 20,000$.

To further discuss the normalized f values shown in Figure 9b at $Re = 20,000$, additional images are provided. Just as the side views of two-dimensional velocity plots on the selected zx planes were used to explain the behavior of the normalized f at $Re = 10,000$ for the central-hole configuration in Figure 13a–c, the velocity plots from the same three zx planes are utilized in Figure 21a–c to explain this behavior at $Re = 20,000$. The structure of a large counterclockwise vortex on the leeward side of the VG is still observed in all three subfigures of Figure 21, suggesting that a similar flow mechanism occurred at $Re = 20,000$ as at $Re = 10,000$. However, at this higher Re , an additional small vortex is found beneath the hole (indicated with a black arrow in Figure 21b). A clearer view of this small vortex is provided by the pathlines plot in Figure 22a. This markedly differs from the flow field for the central-hole configuration at $Re = 10,000$ (Figure 13b). This is because the size of the small vortex varies with Re , similar to the behavior of the wake formed behind a blunt body. Although the wake size can generally increase, decrease, or remain unchanged with Re depending on the flow regime, the simulated results show that it increased with Re under these conditions. Therefore, in this study, as Re increased, the small vortex grew larger and became more easily visible. Consequently, the larger vortex observed in Figure 21b indicates significantly lower pressure on the surface where it attaches.

The velocity plots for the dual-edge, leading-edge, and trailing-edge-hole configurations (Figures 23–25) are nearly identical to those shown in Figures 14–16. Specifically, small vortices (indicated with black arrows) are only found for dual-edge-hole and leading-edge-hole configurations (Figures 23c and 24c, respectively); this is not observed in Figures 23a–b and 24a–b, or any subfigures of the trailing-edge-hole configuration (Figure 25a–c). Clearer views of these small vortices are provided by the pathlines plots in Figure 22b,c. These figures suggest that the flow field patterns for these three hole configurations at $Re = 20,000$ were analogous to those at $Re = 10,000$, and that Re had an insignificant effect on the small vortex size.

Small vortices create very low pressure on the leeward sides of VGs, resulting in higher pressure drags and f . A small vortex is observed for the central-hole configuration at $Re = 20,000$, indicating a higher f ($f_{\text{Central hole}}$). Consequently, the normalized f ($f/f_{\text{Central hole}}$) values of the other three hole configurations are less than those observed at lower Re as evinced in Figure 9b.

This indicates that the normalized Nu and f at $Re = 20,000$ behave differently from those at lower Re because the flow-field structure for the central-hole configuration is sensitive to the increase in Re ; therefore, this flow field undergoes considerable changes contrary to those of other hole configurations.

However, according to [33–35], the combined effects of both Nu and f can be evaluated using the TEF, defined as follows:

$$TEF = (Nu/Nu_0)(f/f_0)^{-1/3}, \quad (11)$$

where Nu_0 and f_0 are the Nusselt number and friction factor for a smooth duct, respectively. These can be calculated using the Dittus–Boelter equation and the Blasius correlation as follows:

$$Nu_0 = 0.23Re^{0.8}Pr^{0.4}, \quad (12)$$

$$f_0 = 0.316Re^{-0.25}. \quad (13)$$

According to Zhai et al. [35], the TEF represents the ratio between the heat transfer coefficient of a heat exchanger duct with VGs and that of a smooth duct under the condition of equal blowing power. It is therefore a useful parameter for comparing the thermal performance of heat exchanger ducts.

In this study, TEF values at various Re for the four hole configurations are presented in Table 1. The TEF values range from 1.282 to 1.600, indicating that all hole configurations offer better thermal performance than the smooth heat exchanger duct within the considered Re range. The graphical comparison of TEF values among the four configurations is shown in Figure 26, plotted as functions of Re . It is evident that the trailing-edge-hole and central-edge-hole configurations offer comparatively higher TEF values. Specifically, the TEF values for the trailing-edge-hole configuration are higher when $Re \leq 15,000$, while the TEF value for the central-hole configuration is higher at $Re = 20,000$. In contrast, the leading-edge-hole and dual-edge-hole configurations provide comparatively lower TEF values. Notably, the TEF values for the dual-edge-hole configuration are the lowest when $Re \leq 7000$, while the TEF values for the leading-edge-hole configuration are the lowest when $Re \geq 10,000$. Overall, the trailing-edge-hole configuration exhibits the best performance, closely followed by the central-edge-hole configuration, whereas the leading-edge-hole and dual-edge-hole configurations show comparatively lower performance.

Charoensin-O-larn and Lertnuwat [34] studied right trapezoidal-winglet VGs installed in upwind and downwind directions (Figure 27a). Their structures were similar to those

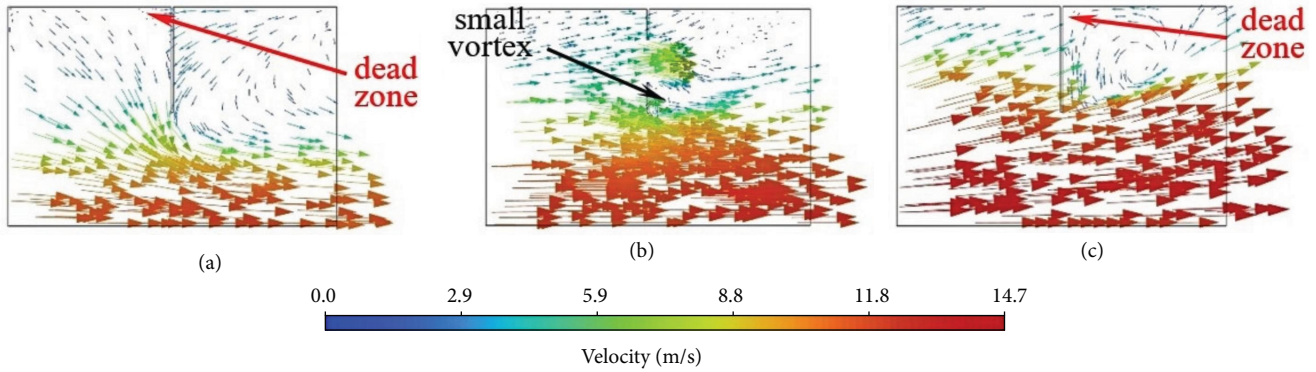


FIGURE 21: Side views of two-dimensional velocity plots on selected zx planes for the central-hole configuration at $Re = 20,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

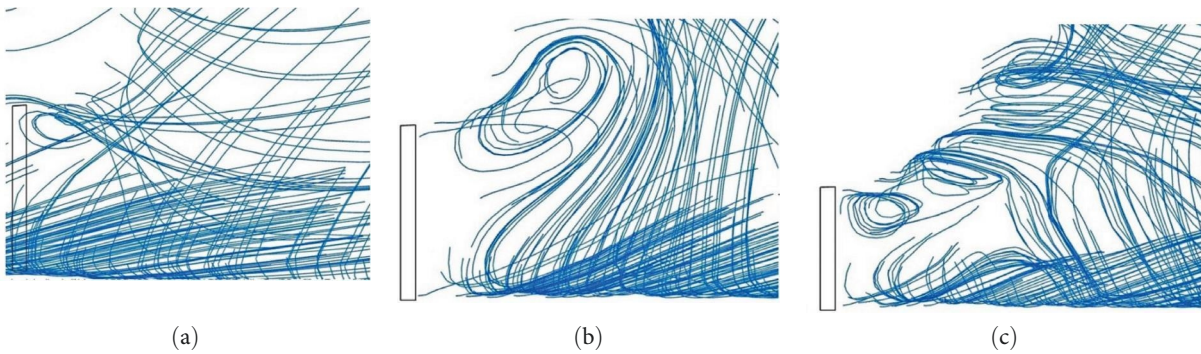


FIGURE 22: Side views of pathlines on an zx plane illustrating the small vortex structure on the leeward side for the (a) central-hole (at $y = 6.25$ mm), (b) dual-edge-hole (at $y = 12.50$ mm), and (c) leading-edge-hole (at $y = 12.50$ mm) configurations at $Re = 20,000$.

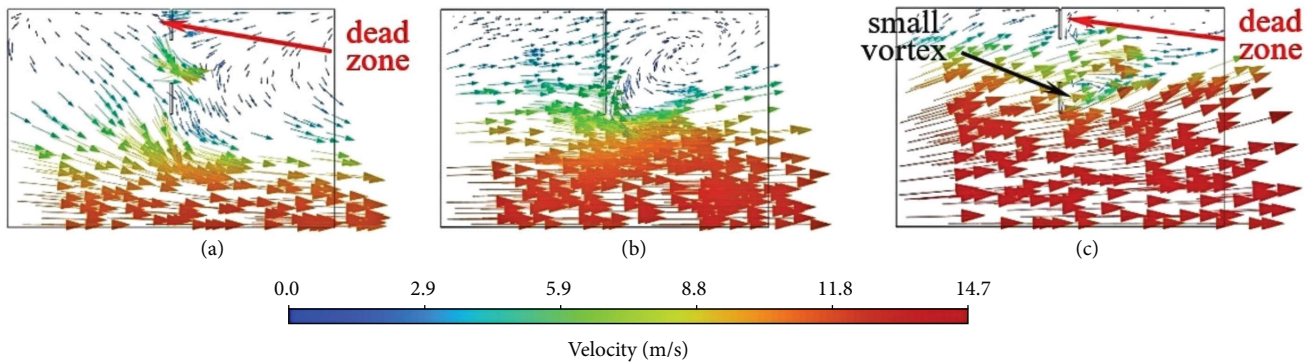


FIGURE 23: Side views of two-dimensional velocity plots on selected zx planes for the dual-edge-hole configuration at $Re = 20,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

of the leading-edge and trailing-edge hole VGs (Figure 27b), respectively; however, thermohydraulic performance differed considerably. At $3000 \leq Re \leq 20,000$, the upwind-direction trapezoidal VG yielded higher TEF compared with the downwind-direction trapezoidal VG. The leading-edge hole VG, in contrast, offered lower TEF compared with the trailing-edge hole VG. Despite their structural similarities, the leading-edge-hole configuration failed to increase TEF as effectively as the upwind-direction trapezoidal VG. The portion of the VG underneath the leading-edge hole likely accounted for this trend in addition to the different blockage ratios.

5. Conclusion

The thermohydraulic performances of rectangular-winglet VGs featuring circular holes on their leading and trailing edges were numerically investigated and compared with that of the central-hole VG at Re ranging from 3000 to 20,000. These findings were categorized based on the Re regimes:

1. Lower Re regime ($3000 \leq Re \leq 15,000$): The Nu values were almost similar across all hole configurations. In contrast, the f values of the leading-edge-hole

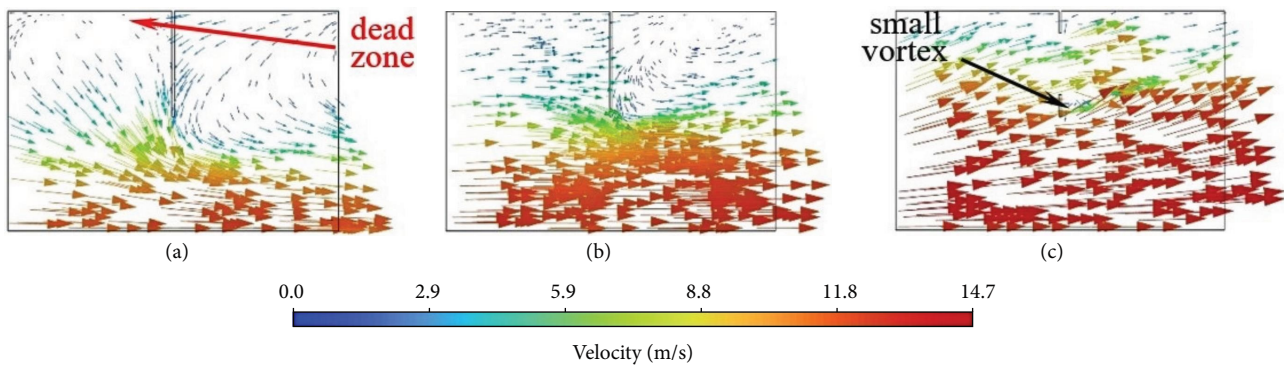


FIGURE 24: Side views of two-dimensional velocity plots on selected zx planes for the leading-edge-hole configuration at $Re = 20,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

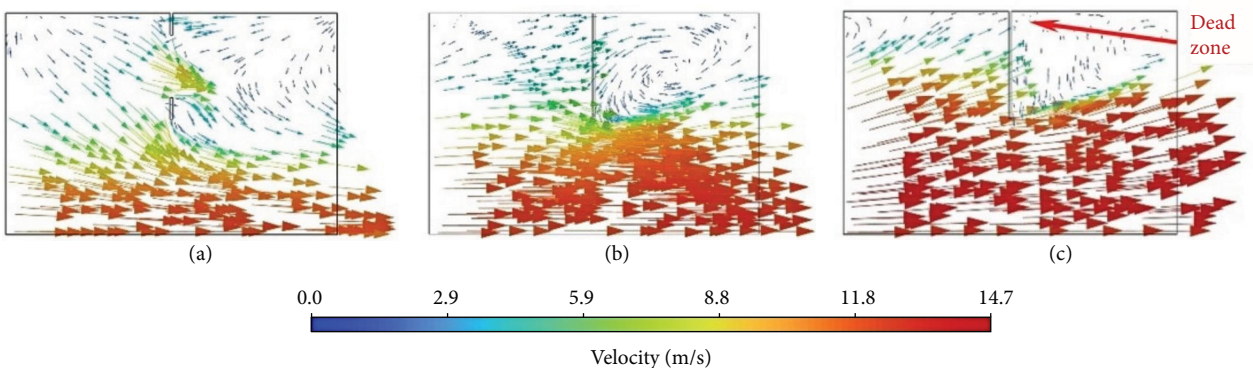


FIGURE 25: Side views of two-dimensional velocity plots on selected zx planes for the trailing-edge-hole configuration at $Re = 20,000$. (a) At $y = 0.00$ mm. (b) At $y = 6.25$ mm. (c) At $y = 12.50$ mm.

TABLE 1: Thermal enhancement factor (TEF) for the four hole configurations.

Re	TEF			
	Central-hole	Dual-edge-hole	Leading-edge-hole	Trailing-edge-hole
3000	1.578	1.541	1.567	1.585
5000	1.499	1.451	1.468	1.502
7000	1.535	1.490	1.501	1.553
10,000	1.545	1.502	1.495	1.574
15,000	1.560	1.519	1.485	1.600
20,000	1.424	1.301	1.282	1.394

and dual-edge-hole configurations were higher than that of the central-hole configuration at the same Re .

- Higher Re regime ($Re = 20,000$): Lower Nu values were observed for the dual-edge- and leading-edge-hole configurations. Only the f value of the leading-edge-hole configuration was considerably higher than that of the central-hole configuration.

However, based on the TEF, the results revealed that, overall, the trailing-edge-hole configuration exhibited superior thermohydraulic performance (TEF: 1.394–1.600), closely followed by the central-hole configuration (TEF: 1.424–1.578),

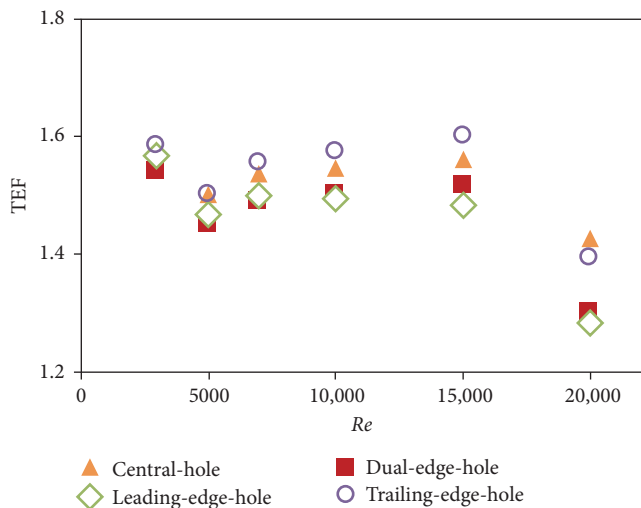


FIGURE 26: Visual comparison of thermal enhancement factor (TEF) for the four hole configurations.

while the leading-edge- and dual-edge-hole configurations showed comparatively lower performance, with TEF ranges of 1.282–1.567 and 1.301–1.541, respectively.

Several velocity plots on the selected zx planes revealed distinct vortex structures, particularly small vortices beneath the leading-edge and central holes. These flow structures were

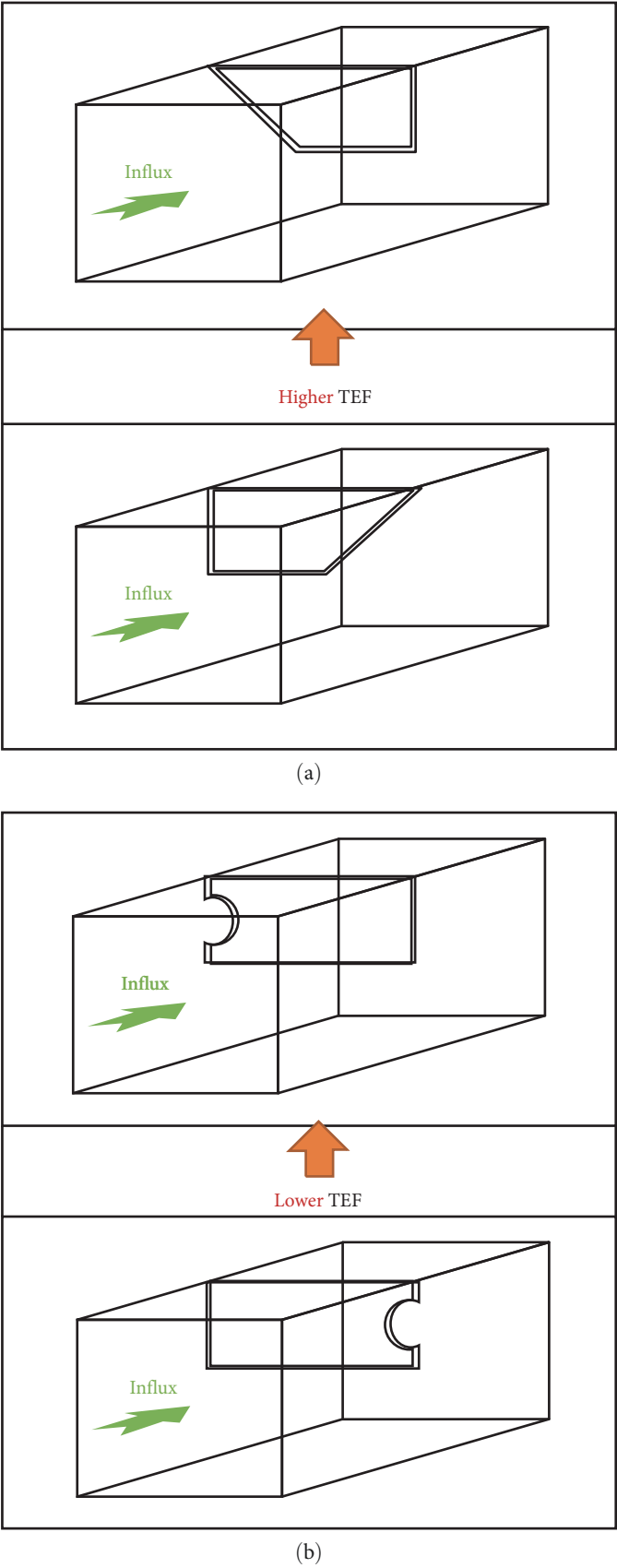


FIGURE 27: Comparison between (a) upwind/downwind-direction trapezoidal vortex generators and (b) leading/trailing-edge-hole vortex generators.

primarily responsible for the differences observed in the thermohydraulic performances across all configurations.

In light of these findings, it can be concluded that the semicircular trailing-edge hole is preferred for the rectangular-winglet VG, as it enhances thermohydraulic performance the most across almost all conditions, except at $Re = 20,000$, where it ranks second to the central hole. However, if difficulties arise, such as aligning two semicircular holes to form a complete circular hole, the central hole serves as a viable alternative, offering nearly equivalent performance. In contrast, the leading-edge and dual-edge holes are less effective and should be avoided.

Nomenclature

Dimensionless Numbers

f :	Friction factor
Nu :	Nusselt number
Pr :	Prandtl number
Re :	Reynolds number
TEF:	Thermal enhancement factor
%diff	Percentage difference

Fluid Properties

C_p :	Specific heat capacity of air
h :	Convection heat transfer coefficient of air
K :	Thermal conductivity of air
T :	Temperature
μ :	Dynamic viscosity of air
ρ :	Density of air

Geometry

A :	Area
c :	Base of winglet vortex generator
D :	Hydraulic diameter
d :	Diameter of punched hole
e :	Winglet vortex generator height
H :	Duct height
L :	Duct length
P_l :	Length of longitudinal pitch
P_t :	Length of transverse pitch
W :	Duct width

Thermohydraulic Parameter

$\dot{m}$:	Rate of mass airflow
q'' :	Heat flux
u :	x -component of velocity
Δp :	Pressure drop

Subscripts

0:	Smooth duct
air:	Air
f :	Friction factor
h :	Hole

i :	Inlet/influx
Nu :	Nusselt number
sur:	Surface of absorber
w :	Winglet vortex generator.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Conflicts of Interest

The authors declare no conflicts of interest.

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